

A Complete Floquet–Jordan Atlas for the Two-Step Hill Oscillator

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Abstract

We classify the periodic two-step Hill oscillator for arbitrary real segment coefficients, including oscillatory, zero-frequency, and hyperbolic pieces. Entire segment functions give one exact $SL(2, \mathbb{R})$ monodromy and a closed Floquet discriminant whose elliptic and hyperbolic regions are complete. At the parabolic boundary we distinguish scalar $\pm I$ from nontrivial Jordan growth, and close every integer iterate and Floquet rate. An independent entire-series checker verifies 900 all-sign transfers.

1 One entire transfer for all segment signs

Fix the periodic equation

$$y'' + k(t)y = 0, \quad k(t) = \begin{cases} k_1, & 0 \leq t < \tau_1, \\ k_2, & \tau_1 \leq t < T, \end{cases} \quad T = \tau_1 + \tau_2 > 0, \quad (1)$$

with $k_j \in \mathbb{R}$ and $\tau_j \geq 0$. Define entire functions

$$C(k, t) = \sum_{m \geq 0} \frac{(-k)^m t^{2m}}{(2m)!}, \quad S(k, t) = \sum_{m \geq 0} \frac{(-k)^m t^{2m+1}}{(2m+1)!}. \quad (2)$$

They obey $C_t = -kS$, $S_t = C$, and $C^2 + kS^2 = 1$. Hence the state $Y = (y, y')^\top$ advances through one constant segment by

$$\Phi(k, t) = \begin{pmatrix} C & S \\ -kS & C \end{pmatrix}, \quad \det \Phi = 1. \quad (3)$$

For $k > 0$, $k = 0$, and $k < 0$, this is respectively the cosine–sine, shear, and cosh–sinh transfer, without a square-root branch convention.

2 Complete stability and boundary theorem

Put $M = \Phi(k_2, \tau_2)\Phi(k_1, \tau_1)$ and write C_j, S_j for the two segment values. Direct multiplication gives

$$\det M = 1, \quad \Delta = \operatorname{tr} M = 2C_1C_2 - (k_1 + k_2)S_1S_2. \quad (4)$$

Theorem 1. *For the oscillator (1):*

1. If $|\Delta| < 2$, the monodromy is elliptic and every solution is bounded. Its Floquet angle per unit time is $T^{-1} \arccos(\Delta/2)$.
2. If $|\Delta| > 2$, the multipliers are a real reciprocal pair. Generic solutions grow exponentially, one forward eigenline decays, and the positive growth rate is $T^{-1} \operatorname{arcosh}(|\Delta|/2)$.
3. If $\Delta = \pm 2$ and $M = \pm I$, every solution is periodic or antiperiodic over one coefficient period. If instead $M \neq \pm I$, then $\operatorname{rank}(M \mp I) = 1$: one Floquet line is bounded and generic solutions grow linearly, with alternating sign on the minus face.

For every $n \geq 1$,

$$M^n = U_{n-1}(\Delta/2)M - U_{n-2}(\Delta/2)I. \quad (5)$$

Here $U_{-1} = 0$ and $U_0 = 1$, so (5) includes $n = 1$.

Proof. The characteristic polynomial is $\lambda^2 - \Delta\lambda + 1$. Its roots are distinct unit conjugates for $|\Delta| < 2$ and distinct real reciprocals for $|\Delta| > 2$, proving the first two cases. At trace ± 2 , a real 2×2 determinant-one matrix is either $\pm I$ or has one Jordan block; its powers distinguish bounded from linear growth. Cayley–Hamilton and the recurrence $U_n(x) = 2xU_{n-1}(x) - U_{n-2}(x)$ prove (5).

3 Exact witnesses and independent receipt

Trace alone does not settle the parabolic dynamics. Four exact sentinels are

$$I, \quad -I, \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} -1 & -1 \\ 0 & -1 \end{pmatrix};$$

the last two have generic linear growth. A hyperbolic exact control is

$$\Phi(-1, \log 2) = \begin{pmatrix} 5/4 & 3/4 \\ 3/4 & 5/4 \end{pmatrix},$$

with multipliers 2 and $1/2$.

The producer evaluates all $6^2 \cdot 5^2 = 900$ parameter rows with $k_j \in \{-4, -1, 0, 1, 4, 9\}$ and $\tau_j \in \{0, 1/4, 1/2, 1, 3/2\}$. The checker does not call its trigonometric or hyperbolic implementation; it rebuilds C, S from 90-term power series, then tests determinant, (4), classification, and every power through $n = 12$.

gate	count	result
all-sign transfer rows	900	19,849 independent assertions
exact boundary matrices	6	scalar/Jordan/hyperbolic PASS
SymPy identities	289	determinant/trace/Chebyshev PASS
hostile mutations	41	41 rejected

Finite rows validate implementation and boundary conventions; the continuum classification follows from Theorem 1.

4 Conclusion and limitations

The all-sign entire transfer, one discriminant, the scalar/Jordan split, and the Chebyshev law together close the two-step family. This does not classify arbitrary periodic coefficients, nonlinear perturbations, or every initial condition’s forward behavior in the hyperbolic case.

References

- [1] G. W. Hill, “On the part of the motion of the lunar perigee which is a function of the mean motions of the sun and moon,” *Acta Mathematica* 8 (1886), 1–36, doi:10.1007/BF02417081.
- [2] Y. F. Golubev, “Resonances of linear differential equations with piecewise constant coefficients,” Keldysh Institute Preprint 43 (1997), MathNet record.